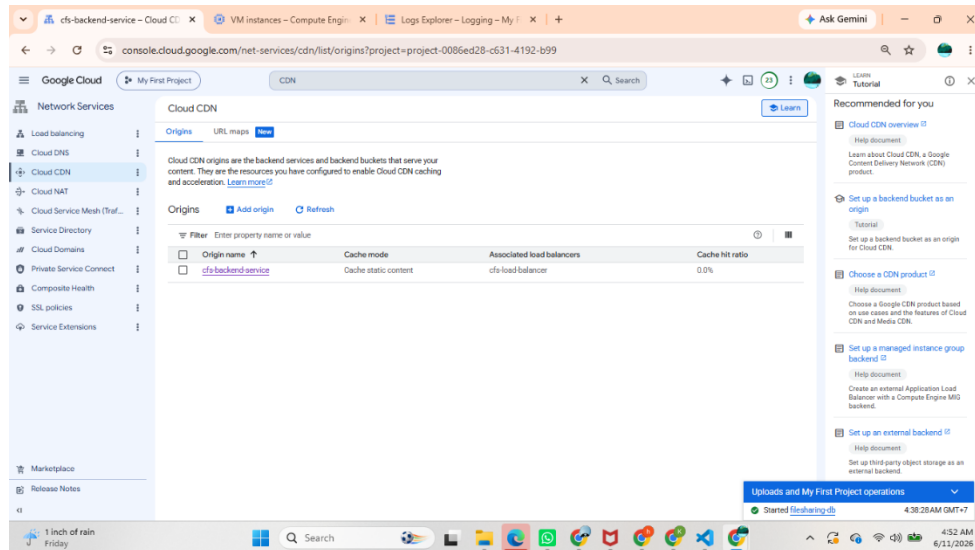


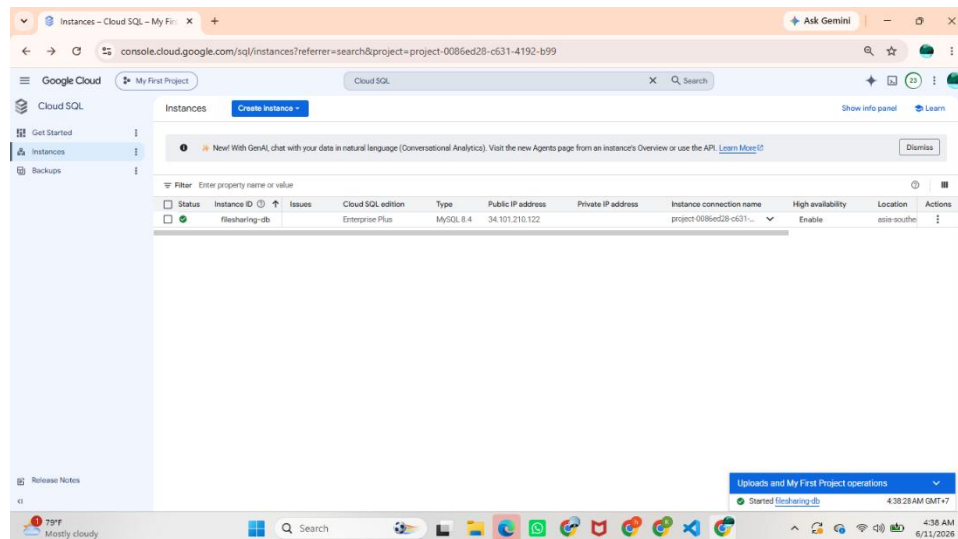
Dokumen Perencanaan Proyek Cloud Computing Minggu 3

A. CDN configuration



CDN (Content Delivery Network) dikonfigurasi menggunakan layanan Cloud CDN milik Google Cloud Platform. Origin yang digunakan adalah cfs-backend-service dengan cache mode Cache static content, yang terhubung langsung ke load balancer cfs-load-balancer. Konfigurasi ini bertujuan untuk mempercepat distribusi konten statis kepada pengguna.

B. Cloud SQL instance



Database dikonfigurasi menggunakan Cloud SQL dengan instance bernama filesharing-db. Instance ini menggunakan MySQL 8.4 dengan edisi Enterprise Plus dan berlokasi di region

asia-southeast2. Instance memiliki Public IP 34.101.210.122 dan fitur High Availability telah diaktifkan untuk memastikan ketersediaan database tetap terjaga.

C. Cloud Storage bucket

The first screenshot shows the 'Buckets' page in the Google Cloud console. A table lists the bucket 'cfs-bucket-kelompok12' with the following details:

Name	Created	Location type	Location	Default storage class	Last modified	Public access	Access control	Protection
cfs-bucket-kelompok12	May 4, 2026, 9:55:32 PM	Region	asia-southeast2	Standard	Jun 1, 2026, 5:00:10 PM	Not public	Uniform	Soft Delete

The second screenshot shows the 'Bucket details' page for 'cfs-bucket-kelompok12'. The 'Objects' tab is selected, showing a list of files in the 'uploads/' folder:

Name	Size	Type	Created
1779448360114-test-final.txt	122 B	text/plain	May 22, 2026, 6:12:40
1779453674592-test-lokal.txt	13 B	text/plain	May 22, 2026, 7:41:14
1780412630584-Materi_Minggu4...	948.8 KB	application/vnd.openxmlformats-officedocument.presentationml.presentation	Jun 2, 2026, 10:03:50
1780412652505-Final_Project_Clo...	209.7 KB	application/pdf	Jun 2, 2026, 10:04:12
1780415398192-activity2_page-00...	245.9 KB	image/jpeg	Jun 2, 2026, 10:49:58
1780415439503-code_3.png	213.6 KB	image/png	Jun 2, 2026, 10:50:39
1780417192973-basket.jpg	318.5 KB	image/jpeg	Jun 2, 2026, 11:19:53

Penyimpanan objek dikonfigurasi menggunakan Cloud Storage dengan bucket bernama cfs-bucket-kelompok12. Bucket berlokasi di region asia-southeast2 (Jakarta) dengan storage class Standard. Public access diatur sebagai Not public untuk menjaga keamanan data, dan fitur Soft Delete serta Object Versioning diaktifkan sebagai perlindungan data. Bucket memiliki folder uploads/ yang digunakan untuk menyimpan file yang diunggah oleh pengguna.

D. Functional Test Report



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Windows\system32> curl.exe -X POST http://8.233.47.193/upload -F "file=@C:\test-success.txt"
{"success":true,"message":"File berhasil diupload","data":{"id":"55804575-f8dd-4a52-b1f9-9796e387a091","filename":"test-success.txt","size":64,"mimeType":"text/plain","gesuri":"gs://cfs-bucket-kelompok12/uploads/1779461348275-test-success.txt","uploadedAt":"2026-05-22T14:49:08.468Z"}}
PS C:\Windows\system32> curl.exe http://8.233.47.193/files
{"success":true,"count":2,"data":[{"id":"55804575-f8dd-4a52-b1f9-9796e387a091","original_name":"test-success.txt","file_size":64,"mime_type":"text/plain","uploaded_at":"2026-05-22T14:49:08.000Z"},{"id":"d1abd7ae-a95e-42f6-be71-b3c9dac35db6","original_name":"test-lokal.txt","file_size":13,"mime_type":"text/plain","uploaded_at":"2026-05-22T12:41:14.000Z"}]}
PS C:\Windows\system32> curl.exe http://8.233.47.193/health
{"status":"healthy","timestamp":"2026-05-22T15:03:27.638Z","hostname":"cloud-file-sharing-vm-web-1","uptime":975.161257616,"memory":{"rss":92499968,"heapTotal":33017856,"heapUsed":30927600,"external":3129790,"arrayBuffers":204073}}
PS C:\Windows\system32>

Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

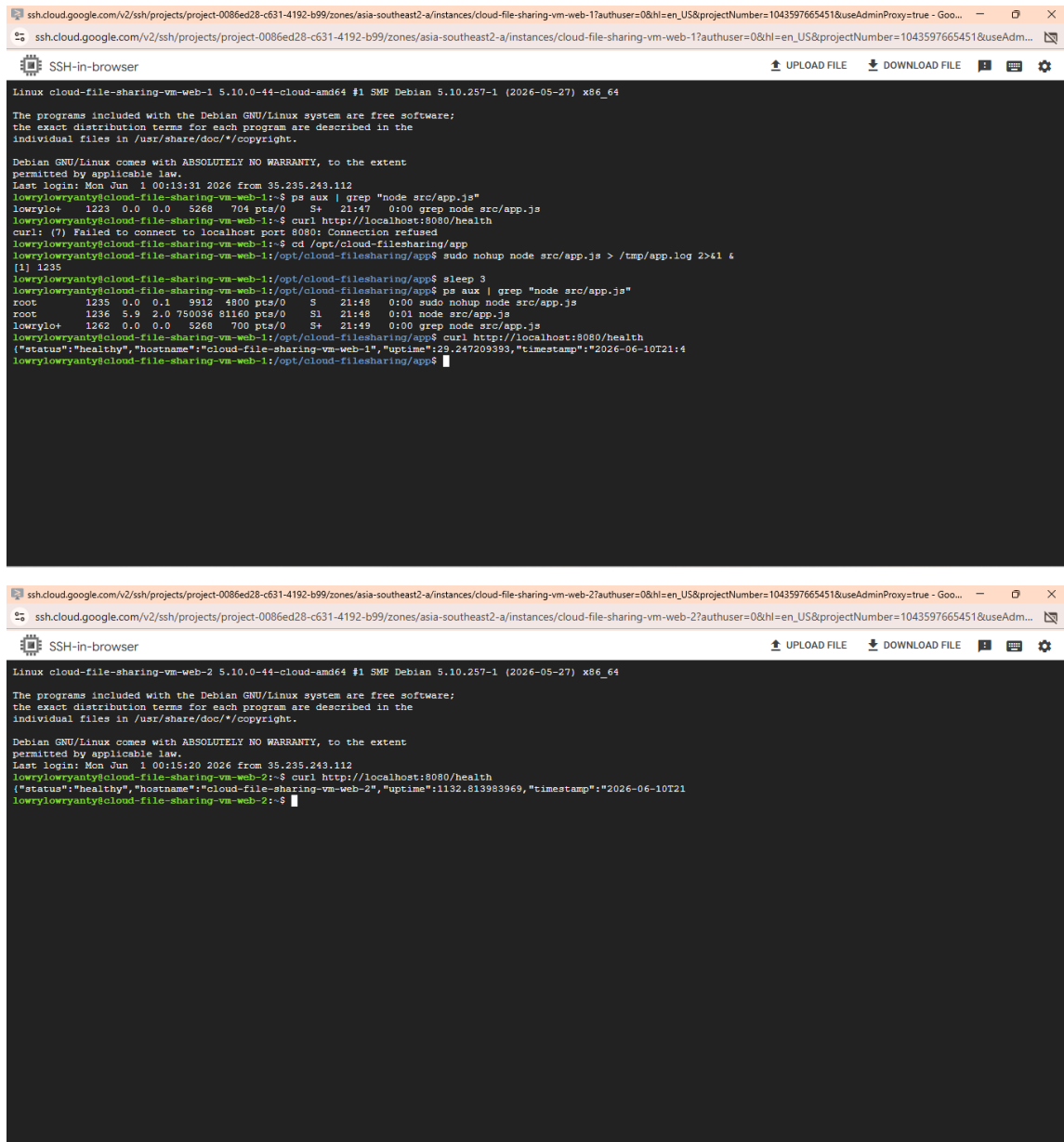
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Windows\system32> curl.exe -X DELETE http://8.233.47.193/file/55804575-f8dd-4a52-b1f9-9796e387a091
{"success":true,"message":"File berhasil dihapus"}
PS C:\Windows\system32>
```

Pengujian fungsional dilakukan menggunakan command-line tools (curl dan PowerShell) untuk memverifikasi empat endpoint utama aplikasi. Hasil pengujian menunjukkan status sukses (200 OK) pada semua skenario: upload file berhasil tersimpan di Cloud Storage dengan

metadata tercatat di database, list files mengembalikan array JSON yang valid, health check merespons dengan status “healthy”, serta delete file berhasil menghapus data dari storage dan database.

E. Health check endpoint



```
Linux cloud-file-sharing-vm-web-1 5.10.0-44-cloud-amd64 #1 SMP Debian 5.10.257-1 (2026-05-27) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Mon Jun 1 00:13:31 2026 from 35.235.243.112
lowrylowryanty@cloud-file-sharing-vm-web-1:~$ ps aux | grep "node src/app.js"
lowrylo 1223 0.0 0.0 5268 704 pts/0 S+ 21:47 0:00 grep node src/app.js
lowrylowryanty@cloud-file-sharing-vm-web-1:~$ curl http://localhost:8080/health
curl: (7) Failed to connect to localhost port 8080: Connection refused
lowrylowryanty@cloud-file-sharing-vm-web-1:~$ cd /opt/cloud-filesharing/app
lowrylowryanty@cloud-file-sharing-vm-web-1:/opt/cloud-filesharing/app$ sudo nohup node src/app.js & /tmp/app.log 2>&1 &
[1] 1235
lowrylowryanty@cloud-file-sharing-vm-web-1:/opt/cloud-filesharing/app$ sleep 3
lowrylowryanty@cloud-file-sharing-vm-web-1:/opt/cloud-filesharing/app$ ps aux | grep "node src/app.js"
root 1235 0.0 0.1 5312 4800 pts/0 S 21:48 0:00 sudo nohup node src/app.js
root 1236 5.9 2.0 750036 81160 pts/0 Sl 21:48 0:01 node src/app.js
lowrylo 1262 0.0 0.0 5268 700 pts/0 S+ 21:49 0:00 grep node src/app.js
lowrylowryanty@cloud-file-sharing-vm-web-1:/opt/cloud-filesharing/app$ curl http://localhost:8080/health
{"status":"healthy","hostname":"cloud-file-sharing-vm-web-1","uptime":29.247209393,"timestamp":"2026-06-10T21:4
lowrylowryanty@cloud-file-sharing-vm-web-1:/opt/cloud-filesharing/app$
```

```
Linux cloud-file-sharing-vm-web-2 5.10.0-44-cloud-amd64 #1 SMP Debian 5.10.257-1 (2026-05-27) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Mon Jun 1 00:15:20 2026 from 35.235.243.112
lowrylowryanty@cloud-file-sharing-vm-web-2:~$ curl http://localhost:8080/health
{"status":"healthy","hostname":"cloud-file-sharing-vm-web-2","uptime":1132.813983969,"timestamp":"2026-06-10T21
lowrylowryanty@cloud-file-sharing-vm-web-2:~$
```

Endpoint /health diimplementasikan pada port 8080 untuk memantau ketersediaan aplikasi secara real-time melalui metode HTTP GET. Endpoint ini mengembalikan respons JSON yang berisi status server (“healthy”), hostname instans (bergantian antara web-1 dan web-2), uptime proses, dan timestamp terkini.

F. Load Balancer configuration

The screenshot displays the Google Cloud console interface for configuring a Load Balancer. The top panel shows the 'Load balancing' overview, and the bottom panel shows the 'Load balancer details' for the 'cfs-load-balancer'.

Load balancing overview:

- Monitor + Analyze Load Balanced Traffic:** Use VPC Flow Logs and Flow Analyzer to gain flow level insights for your Load Balancers and backends. [Learn more](#)
- ✓ Diagnose connectivity issues
- ✓ Identify top talkers + Analyze network utilization
- ✓ For passthrough load balancers, understand how flows are being distributed among their backends
- [Try now](#) | [Remind me later](#)

Load balancers:

Name	Load balancer type	Access type	Protocols	Region	Backends	Actions
cfs-load-balancer	Application	External	HTTP		1 backend service (1 instance group, 0 network endpoint groups)	

To view or delete load balancing resources like forwarding rules and target proxies, go to the [load balancing components view](#).

Load balancer details:

Routing rules

Hosts	Paths	Backend
All unmatched (default)	All unmatched (default)	cfs-backend-service

Backend

Backend services

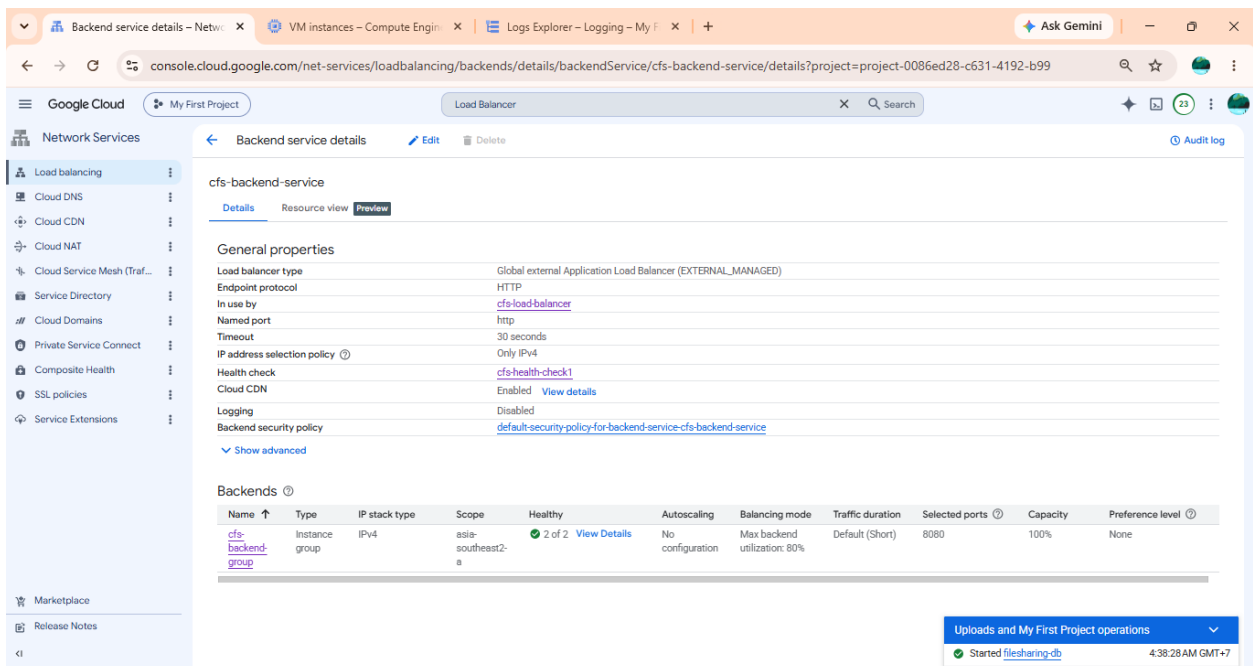
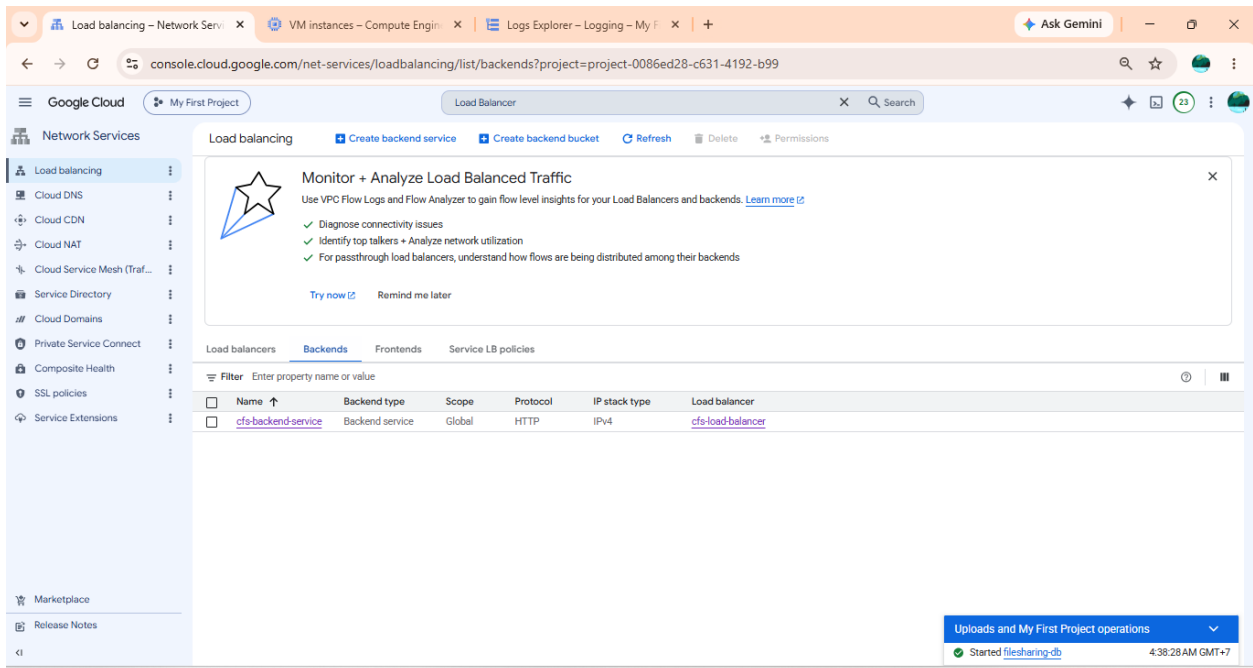
1. cfs-backend-service

Endpoint protocol	HTTP
Named port	http
Timeout	30 seconds
IP address selection policy	Only IPv4
Health check	cfs-health-check1
Cloud CDN	Enabled View details
Logging	Disabled
Backend security policy	default-security-policy-for-backend-service-cfs-backend-service

[Show advanced](#)

Backends

Name	Type	IP stack type	Scope	Healthy	Autoscaling	Balancing mode	Traffic duration	Selected ports	Capacity	Preference level
cfs-backend-group	Instance group	IPv4	asia-southeast2-a	2 of 2 View Details	No configuration	Max backend utilization: 80%	Default (Short)	8080	100%	None



Load Balancer dikonfigurasi menggunakan Global External Application Load Balancer dengan nama cfs-load-balancer yang mendengarkan traffic HTTP pada port 80 via IP publik 8.233.47.193. Backend service cfs-backend-service terhubung ke instance group dengan dua VM (web-1 dan web-2) melalui port 8080, menggunakan mode balancing berbasis utilization (target 80%). Integrasi dengan health check cfs-health-check1 memastikan distribusi traffic yang merata dan otomatis failover ke instans sehat jika terjadi kegagalan.